

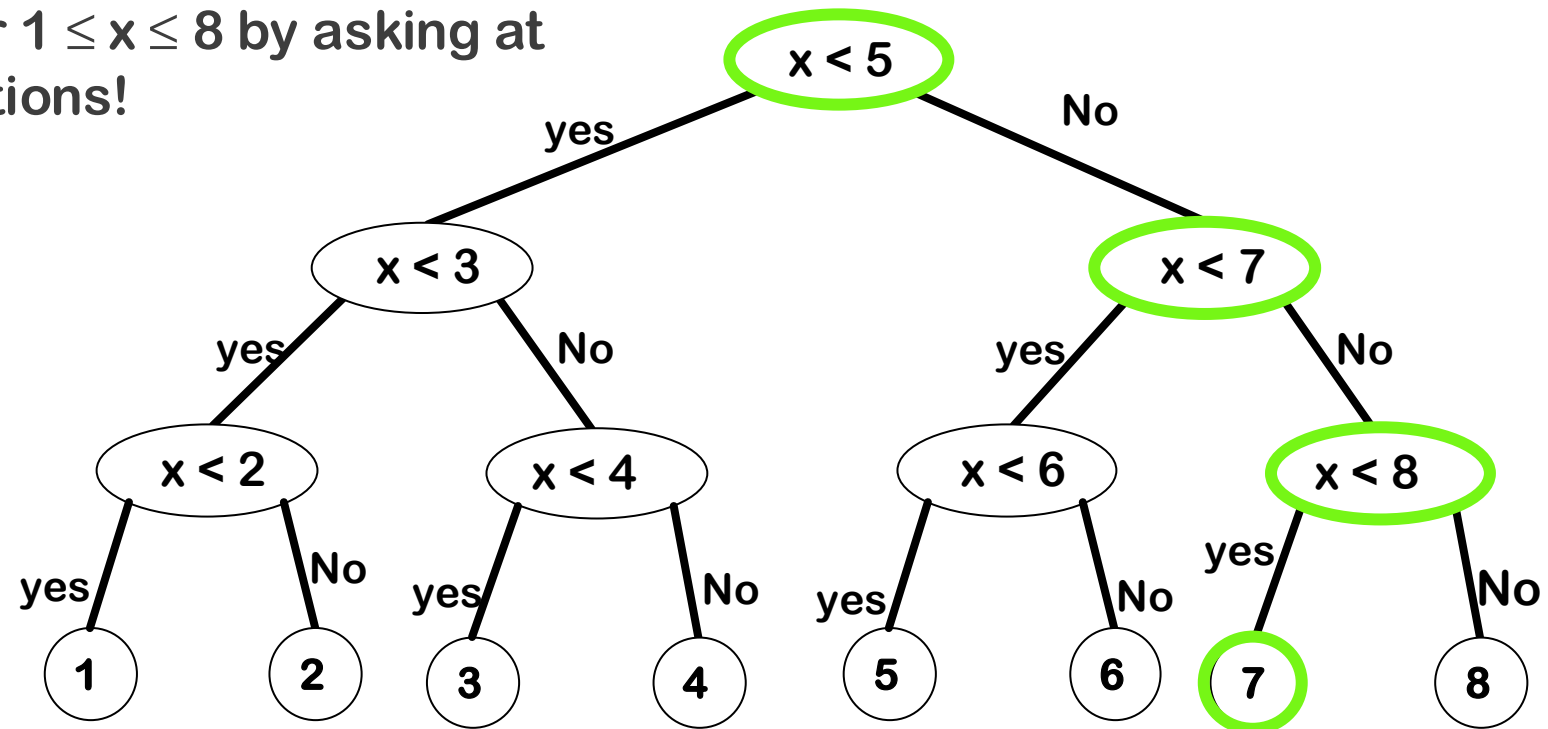


DIVIDE AND CONQUER

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NJIT

GUESS A NUMBER!

- Guess an integer $1 \leq x \leq 8$ by asking at most three questions!



GUESS A NUMBER!

- Guess an integer $1 \leq x \leq 16$ by asking at most three questions!
- Guess 7

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

$x < 9??$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

$x < 5??$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

$x < 7??$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

$x < 8??$

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----

INTRODUCTION



Napoleon



DEFINITION

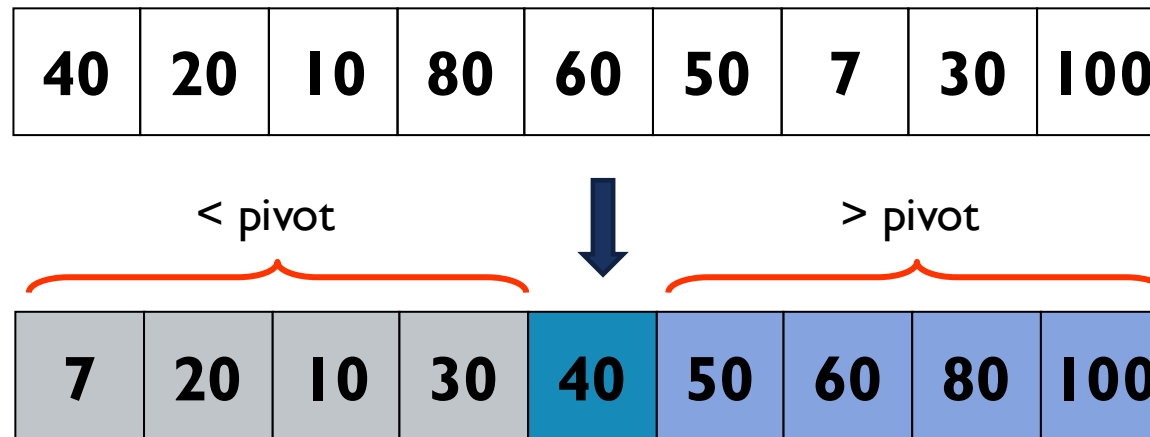
- **Divide** the problem into a number of subproblems that are smaller instances of the same problem.
- **Conquer** the subproblems by solving them recursively. If they are small enough, solve the subproblems as base cases.
- **Combine** the solutions to the subproblems into the solution for the original problem.

EXAMPLE 1: QUICKSORT

- **Divide** the list into two smaller list: less than pivot and greater than pivot.
- **Conquer** the subproblems by sorting the sublists recursively.
- **Combine** the sorted sublists to obtain the sorted main list.

QUICKSORT: DIVIDE

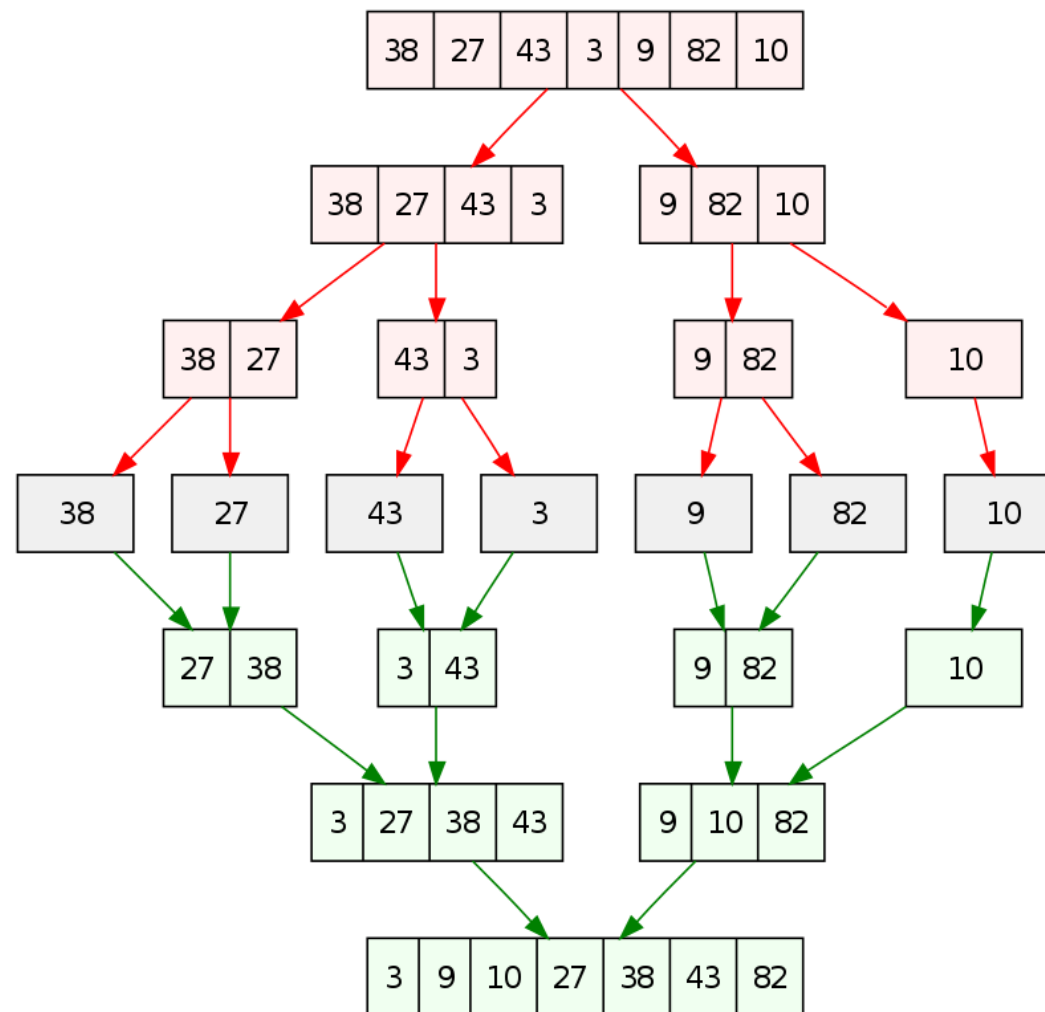
- **Divide** the list into two smaller list: less than pivot and greater than pivot.



EXAMPLE 2: MERGESORT

- **Divide** the list into two equal size sublists.
- **Conquer** the subproblems by sorting the lists recursively.
- **Combine(Merge)** the sorted lists to obtain the sorted list.

MERGESORT



EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} 7 & 1 & 4 & 5 & 6 & 7 & 8 & 9 \\ 1 & 2 & 4 & 6 & 3 & 5 & 8 & 3 \\ 5 & 3 & 6 & 7 & 4 & 1 & 3 & 2 \\ 0 & 4 & 0 & 6 & 7 & 5 & 7 & 2 \\ 0 & 4 & 6 & 7 & 0 & 1 & 6 & 3 \\ 8 & 9 & 4 & 6 & 3 & 5 & 8 & 4 \\ 4 & 6 & 6 & 7 & 4 & 1 & 3 & 6 \\ 2 & 0 & 0 & 6 & 7 & 5 & 7 & 0 \end{bmatrix}_{n \times n}$$

n^2

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{bmatrix} 7 & 5 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 4 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 7+1 & 5+4 \\ 3+8 & 2+6 \end{bmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		11	2	7	7	10	7	14	16
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		8	7	11	8	10	11	15	6
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		5	4	12	10	4	5	9	5
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4		3	9	8	10	10	11	14	6
0	4	6	7	0	1	6	3	+	4	1	3	6	4	0	6	7	=	4	5	9	13	4	1	12	10
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		15	14	11	6	10	11	15	4
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		4	12	13	7	8	1	9	13
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		8	7	0	12	14	11	14	0

n^2

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

The diagram illustrates the first step of Strassen's matrix multiplication. It shows the multiplication of two 2x2 matrices. The first matrix has elements 7 (red), 5 (red), 3 (blue), and 2 (blue). The second matrix has elements 1 (black), 4 (green), 8 (black), and 6 (green). An orange box highlights the top row of the first matrix [7, 5] and the first column of the second matrix [1, 8]. An orange line with an arrow connects this box to the top-left element of the result matrix, which is the expression 7x1+5x8. The other elements of the result matrix are 7x4+5x6, 3x1+2x8, and 3x4+2x6, with their respective terms color-coded to match the input matrices.

$$\begin{bmatrix} 7 & 5 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 7 \times 1 + 5 \times 8 & 7 \times 4 + 5 \times 6 \\ 3 \times 1 + 2 \times 8 & 3 \times 4 + 2 \times 6 \end{bmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

The diagram illustrates the calculation of the top-right element of the product matrix in Strassen's algorithm. It shows the multiplication of two 2x2 matrices:

$$\begin{bmatrix} 7 & 5 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 4 \\ 8 & 6 \end{bmatrix} = \begin{bmatrix} 7 \times 1 + 5 \times 8 & 7 \times 4 + 5 \times 6 \\ 3 \times 1 + 2 \times 8 & 3 \times 4 + 2 \times 6 \end{bmatrix}$$

An orange line connects the top row of the first matrix (7, 5) to the top-right element of the product matrix (7x4 + 5x6). Another orange line connects the second column of the second matrix (4, 6) to the same top-right element. The elements 7, 5, 4, and 6 are highlighted with orange boxes, and the corresponding terms in the product matrix are also highlighted with orange boxes.

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7	=	74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4	=	121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7		74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7	=	74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

$$n - 1 \rightarrow n^2(n - 1) = O(n^3)$$

EXAMPLE 3: MATRIX MULTIPLICATION

$$\mathbf{C}_{n \times n} = \mathbf{A}_{n \times n} \times \mathbf{B}_{n \times n}$$

```
for (i=0; i<n; i++)  
    for (j=0; j<n; j++)  
    {  
        C[i][j]=0;  
        for (k=0; k<n; k++)  
            C[i][j] += A[i][k] * B[k][j];  
    }
```

$$O(n^3) \rightarrow O(n^?)$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9	4	1	3	2	4	0	6	7
1	2	4	6	3	5	8	3	7	5	7	2	7	6	7	0
5	3	6	7	4	1	3	2	0	1	6	3	0	4	6	3
0	4	0	6	7	5	7	2	3	5	8	4	3	6	7	4
0	4	6	7	0	1	6	3	4	1	3	6	4	0	6	7
8	9	4	6	3	5	8	4	7	5	7	0	7	6	7	0
4	6	6	7	4	1	3	6	0	6	7	0	4	0	6	7
2	0	0	6	7	5	7	0	6	7	0	6	7	6	7	0

$\times$

$= ?$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7	=	74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9	4	1	3	2	4	0	6	7	=	177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3	7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2	0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2	3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7	74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0	184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7	138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0	89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9	4	1	3	2	4	0	6	7	=	177	193	215	138	218	148	304	179	
1	2	4	6	3	5	8	3	7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120	
5	3	6	7	4	1	3	2	0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130	
0	4	0	6	7	5	7	2	3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122	
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7		74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9		4	1	3	2	4	0	6	7		177	193	215	138	218	148	304	179
1	2	4	6	3	5	8	3		7	5	7	2	7	6	7	0		101	142	189	78	136	112	208	120
5	3	6	7	4	1	3	2		0	1	6	3	0	4	6	3		97	102	168	98	111	102	199	130
0	4	0	6	7	5	7	2		3	5	8	4	3	6	7	4		121	138	181	86	151	102	203	122
0	4	6	7	0	1	6	3	x	4	1	3	6	4	0	6	7	=	74	123	169	72	101	114	177	88
8	9	4	6	3	5	8	4		7	5	7	0	7	6	7	0		184	191	259	112	220	160	306	169
4	6	6	7	4	1	3	6		0	6	7	0	4	0	6	7		138	144	186	126	156	144	242	123
2	0	0	6	7	5	7	0		6	7	0	6	7	6	7	0		89	106	159	70	117	66	173	136

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$$

$$\begin{aligned} P_1 + P_2 &= \\ A(F-H) + (A+B)H &= \\ AF - AH + AH + BH &= \\ AF + BH \end{aligned}$$

Trick:

$$P_1 = A \cdot (F-H)$$

$$P_2 = (A+B) \cdot H$$

$$P_3 = (C+D) \cdot E$$

$$P_4 = D \cdot (G-E)$$

$$P_5 = (A+D) \cdot (E+F)$$

$$P_6 = (B-D) \cdot (G+H)$$

$$P_7 = (A-C) \cdot (E+F)$$

$$AE+BG = P_5 + P_4 - P_2 + P_6$$

$$AF+BH = P_1 + P_2$$

$$CE+DG = P_3 + P_4$$

$$CF+DH = P_5 + P_1 - P_3 - P_7$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} AE+BG & AF+BH \\ CE+DG & CF+DH \end{pmatrix}$$

Trick:

$$\begin{aligned} P_1 &= A \cdot (F-H) & P_5 &= (A+D) \cdot (E+H) \\ P_2 &= (A+B) \cdot H & P_6 &= (B-D) \cdot (G+H) \\ P_3 &= (C+D) \cdot E & P_7 &= (A-C) \cdot (E+F) \\ P_4 &= D \cdot (G-E) \end{aligned}$$

$$\begin{aligned} AE+BG &= P_5 + P_4 - P_2 + P_6 \\ AF+BH &= P_1 + P_2 \\ CE+DG &= P_3 + P_4 \\ CF+DH &= P_5 + P_1 - P_3 - P_7 \end{aligned}$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \times \begin{pmatrix} E & F \\ G & H \end{pmatrix} = \begin{pmatrix} P_5 + P_4 - P_2 + P_6 & P_1 + P_2 \\ P_3 + P_4 & P_5 + P_1 - P_3 - P_7 \end{pmatrix}$$

Trick:

$$P_1 = A \cdot (F - H)$$

$$P_2 = (A + B) \cdot H$$

$$P_3 = (C + D) \cdot E$$

$$P_4 = D \cdot (G - E)$$

$$P_5 = (A + D) \cdot (E + H)$$

$$P_6 = (B - D) \cdot (G + H)$$

$$P_7 = (A - C) \cdot (E + F)$$

$$AE + BG = P_5 + P_4 - P_2 + P_6$$

$$AF + BH = P_1 + P_2$$

$$CE + DG = P_3 + P_4$$

$$CF + DH = P_5 + P_1 - P_3 - P_7$$

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

7	1	4	5	6	7	8	9	4	1	3	2	4	0	6	7	
1	2	4	6	3	5	8	3	7	5	7	2	7	6	7	0	
5	3	6	7	4	1	3	2	0	1	6	3	0	4	6	3	
0	4	0	6	7	5	7	2	3	5	8	4	3	6	7	4	
0	4	6	7	0	1	6	3	4	1	3	6	4	0	6	7	=?
8	9	4	6	3	5	8	4	7	5	7	0	7	6	7	0	
4	6	6	7	4	1	3	6	0	6	7	0	4	0	6	7	
2	0	0	6	7	5	7	0	6	7	0	6	7	6	7	0	

EXAMPLE 3: STRASSEN'S MATRIX MULTIPLICATION

Matrix Strassen(A, B, n) :

```
if (n==1) return A*B
```

```
a, b, c, d = split(A)
```

```
e, f, g, h = split(B)
```

```
p1 = Strassen(a, f - h, n/2)
```

```
p2 = Strassen(a + b, h, n/2)
```

```
p3 = Strassen(c + d, e, n/2)
```

```
p4 = Strassen(d, g - e, n/2)
```

```
p5 = Strassen(a + d, e + h, n/2)
```

```
p6 = Strassen(b - d, g + h, n/2)
```

```
p7 = Strassen(a - c, e + f, n/2)
```

```
c11 = p5 + p4 - p2 + p6;
```

```
c12 = p1 + p2
```

```
c21 = p3 + p4
```

```
c22 = p1 + p5 - p3 - p7
```

```
return C
```

Matrix Strassen(A, B, n) $\longrightarrow T(n)$
 if (n==1) return A*B $\longrightarrow T(n) = 1$
 a, b, c, d = split(A) $\longrightarrow O(n^2)$
 e, f, g, h = split(B)
 p1 = Strassen(a, f - h, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p2 = Strassen(a + b, h, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p3 = Strassen(c + d, e, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p4 = Strassen(d, g - e, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p5 = Strassen(a + d, e + h, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p6 = Strassen(b - d, g + h, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 p7 = Strassen(a - c, e + f, n/2) $\longrightarrow T\left(\frac{n}{2}\right) + 2\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 c11 = p5 + p4 - p2 + p6 $\longrightarrow 3\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 c12 = p1 + p2 $\longrightarrow \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 c21 = p3 + p4 $\longrightarrow \left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 c22 = p1 + p5 - p3 - p7 $\longrightarrow 3\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$
 return C

$$T(n) = O(n^2) + 7T\left(\frac{n}{2}\right) + 18\left(\frac{n}{2}\right)\left(\frac{n}{2}\right)$$

STRASSEN: TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n^2) + 7T\left(\frac{n}{2}\right)$$

$$a = 7, b = 2, k = 2, p = 0$$

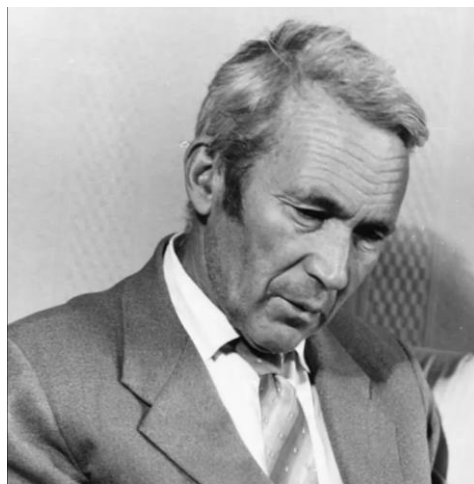
$$T(n) = O(n^{\log_2 7}) = O(n^{2.81})$$

$$T(n) = a T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$$\text{if } a > b^k, \text{ then } T(n) = \theta(n^{\log_b a})$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$\begin{array}{r}
 \\
 \times \begin{matrix} 2 & 5 \\ 6 & 3 \end{matrix} \\
 \hline
 \begin{matrix} 1 & 5 \\ 6 & 0 \\ 3 & 0 \\ 1 & 2 \end{matrix} \\
 + \begin{matrix} 3 & 0 & 0 \\ 1 & 2 & 0 & 0 \end{matrix} \\
 \hline
 \begin{matrix} 1 & 5 & 7 & 5 \end{matrix}
 \end{array}$$



Andrey Kolmogorov

Is in [2] what happened:

In the autumn of 1960 a seminar on problems in cybernetics was held at the Faculty of Mechanics and Mathematics of Moscow University under the direction of Kolmogorov. There Kolmogorov stated the n^2 conjecture and proposed some problems concerning the complexity of the solution of linear systems of equations and some similar calculations. I began to think actively about the n^2 conjecture, and in less than a week I found that the algorithm with which I was planning to derive a lower bound for $M(n)$ gave a bound of the form

$$M(n) = O(n^{\log_2 3}), \quad \log_2 3 = 1.5849 \dots$$



Anatoly A. Karatsuba

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- Storing an n-digit large integer: 8, 256, 734

8	2	5	6	7	3	4
---	---	---	---	---	---	---

- Multiply by a power of 10: 8, 256, 734 $\times 10^4$

8	2	5	6	7	3	4	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- Adding two large integers: **850368000000** + **7519802000** =?

8	5	0	3	6	8	0	0	0	0	0	0
	7	5	1	9	8	0	2	0	0	0	

carry

0

L.I.M: SUMMATION

8	5	0	3	6	8	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

7	5	1	9	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---

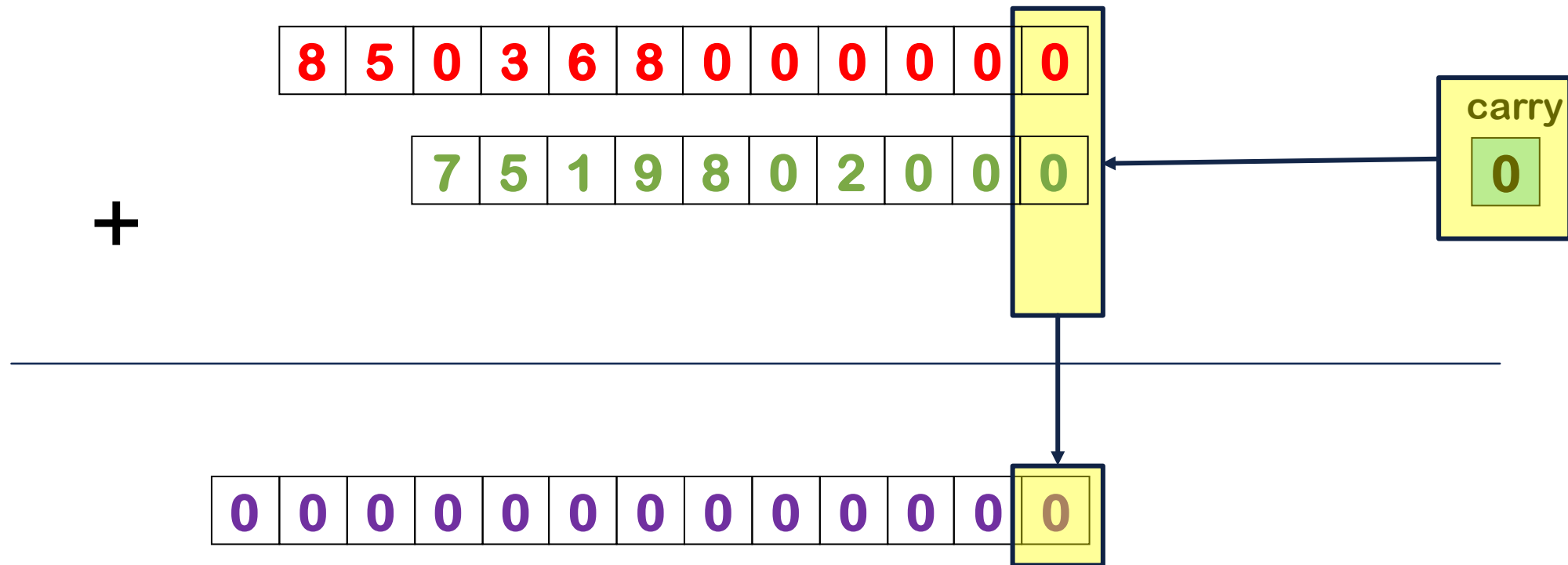
+

carry

0

0	0	0	0	0	0	0	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

L.I.M: SUMMATION



L.I.M: SUMMATION



L.I.M: SUMMATION



L.I.M: SUMMATION





+

8	5	0	3	6	8	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

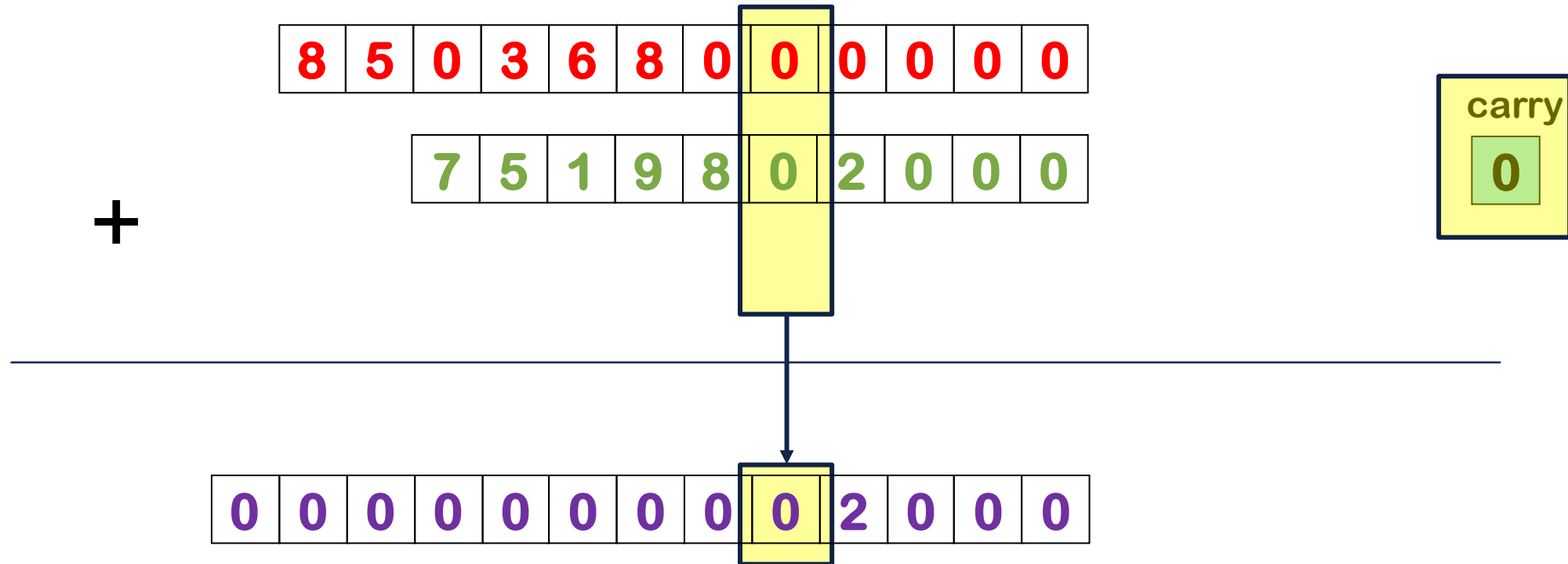
7	5	1	9	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---

carry

1

0	0	0	0	0	8	7	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

L.I.M: SUMMATION



L.I.M: SUMMATION



L.I.M: SUMMATION



L.I.M: SUMMATION

+

8	5	0	3	6	8	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

7	5	1	9	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---

carry

1

0	0	0	0	0	0	7	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

L.I.M: SUMMATION



L.I.M: SUMMATION

+

8	5	0	3	6	8	0	0	0	0	0	0
	7	5	1	9	8	0	2	0	0	0	

carry

0

0	0	0	0	0	8	7	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

L.I.M: SUMMATION



L.I.M: SUMMATION

+

8	5	0	3	6	8	0	0	0	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---

7	5	1	9	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---

carry

0

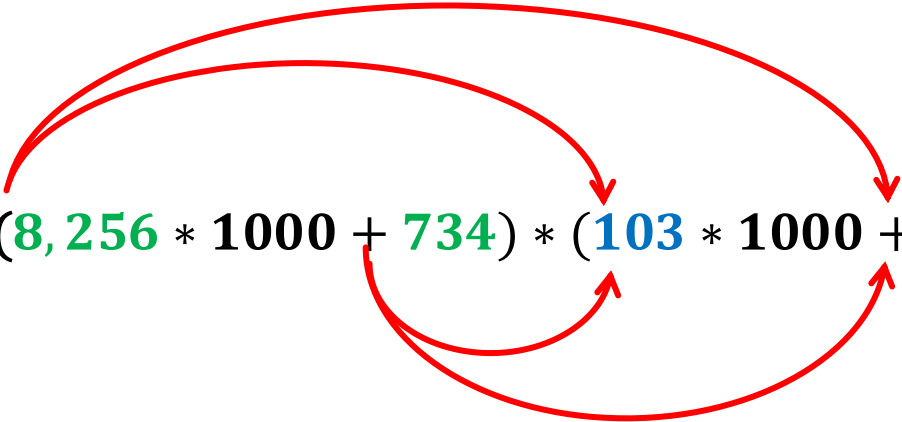
0	8	5	7	8	8	7	8	0	2	0	0	0
---	---	---	---	---	---	---	---	---	---	---	---	---

$O(n)$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8,256 * 1000 + 734) * (103 * 1000 + 875)$$


The diagram illustrates the decomposition of the large integers 8,256,734 and 103,875 into their thousands and remainders. Red curved arrows show the mapping from the original numbers to the terms in the equation. One arrow points from 8,256,734 to 8,256, and another from 734. Similarly, one arrow points from 103,875 to 103, and another from 875. This visualizes the process of splitting a large number into a smaller part multiplied by a power of 10, plus a remainder.

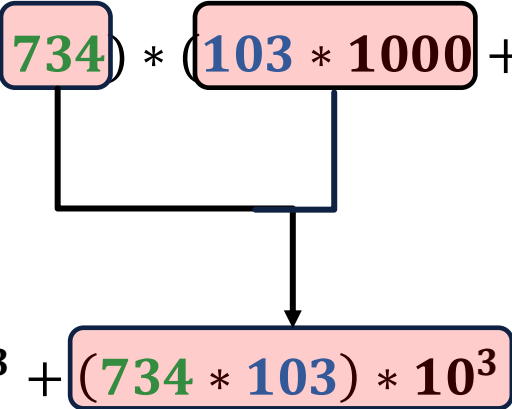
EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$
$$= (8256 * 103) * 10^6 +$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$\begin{aligned} 8,256,734 * 103,875 &= (8256 * 1000 + 734) * (103 * 1000 + 875) \\ &= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + \end{aligned}$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + \boxed{734}) * (\boxed{103 * 1000} + 875)$$

$$= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + (\boxed{734 * 103}) * 10^3 +$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$\begin{aligned} 8,256,734 * 103,875 &= (8256 * 1000 + \boxed{734}) * (103 * 1000 + \boxed{875}) \\ &= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + (\boxed{734 * 103}) * 10^3 + \boxed{734 * 875} \end{aligned}$$

The diagram illustrates the expansion of the multiplication of two large integers. The first line shows the numbers 8,256,734 and 103,875 being split into their thousands and remainder parts. The second line shows the resulting four terms of the expansion, with arrows indicating the mapping from the first line's components to the second line's terms.

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$

$$= (8256 * 103) * 10^6 + (8256 * 875) * 10^3 + (734 * 103) * 10^3 + (734 * 875)$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$

$$= (8256 * 103) * 10^6 + (8256 * 875 + 734 * 103) * 10^3 + (734 * 875)$$

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

$$8,256,734 * 103,875 = (8256 * 1000 + 734) * (103 * 1000 + 875)$$

Not multiplication

$$= (8256 * 103) * 10^6 + (8256 * 875 + 734 * 103) * 10^3 + (734 * 875)$$

4

EXAMPLE 4: LARGE INTEGER MULTIPLICATION

- u and v are two n-digit large integer:

$$u = x * 10^m + y, \quad v = w * 10^m + z \quad m = \left\lceil \frac{n}{2} \right\rceil$$

$$u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^m + y * z$$

Example:

$$7,824,657 \times 64,528 \rightarrow 7824657 \times 0064528 \quad m = 4$$

L.I.M:ALGORITHM

```
multiply(u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

L.I.M:ALGORITHM

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    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

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    t1=multiply(x,z)
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```

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    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
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```

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    t2=multiply(y,w)
    s=sum(t1,t2)
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```

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    for(i=0;i<2*m;i++) p=p+'0'
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    t2=multiply(y,w)
    s=sum(t1,t2)
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```


L.I.M:ALGORITHM

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    for(i=0;i<2*m;i++) p=p+'0'
    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
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```

L.I.M:ALGORITHM

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    t1=multiply(x,z)
    t2=multiply(y,w)
    s=sum(t1,t2)
    for(i=0;i<m;i++) s=s+'0'
    q=multiply(y,z)
    return sum(sum(p,s),q)
```

`multiply(array u,v)`

`if(u=='0' or v=='0') return '0'`

`n1=u.size()`

`n2=v.size()`

`n=max(n1,n2)`

`if(n<threshold) return str(int(u)*int(v))`

`m=n/2`

`x,y=split(u)`

`w,z=split(v)`

`p=multiply(x,w)`

`for(i=0;i<2*m;i++) p=p+'0'`

`t1=multiply(x,z)`

`t2=multiply(y,w)`

`s=sum(t1,t2)`

`for(i=0;i<m;i++) s=s+'0'`

`q=multiply(y,z)`

`return sum(sum(p,s),q)`

$T(n)$

$O(1)$

$O(1)$

$T(1) = 1$

$O(1)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$T\left(\frac{n}{2}\right)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$O(n)$

$T\left(\frac{n}{2}\right)$

$O(n)$

$$T(n) = O(n) + 4T\left(\frac{n}{2}\right)$$

TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n) + 4T\left(\frac{n}{2}\right)$$

$$a = 4, b = 2, k = 1, p = 0$$

$$T(n) = O(n^{\log_2 4}) = O(n^2)$$

$$T(n) = a T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$$\text{if } a > b^k, \text{ then } T(n) = \theta(n^{\log_b a})$$

LIM: MODIFICATION

$$u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^m + y * z$$

LIM: MODIFICATION

$$u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^m + y * z$$

$$x * z + y * w = (x + y) * (z + w) - (x * w) - (y * z)$$

$$(x + y)(z + w) =$$

$$xz + xw + yz + yw =$$

$$(xz + yw) + (xw) + (yz)$$

LIM: MODIFICATION

$$\cancel{u * v = (x * w) * 10^{2m} + (x * z + y * w) * 10^m + y * z}$$

$$x * z + y * w = (x + y) * (z + w) - (x * w) - (y * z)$$

$$r = (x + y) * (z + w), \quad p = x * w, \quad q = y * z$$

$$u * v = p * 10^{2m} + (r - p - q) * 10^m + q$$

L.I.M:ALGORITHM

```
multiply(array u,v)
    if(u=='0' or v=='0') return '0'
    n1=u.size()
    n2=v.size()
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v))
    m=n/2
    x,y=split(u)
    w,z=split(v)
    p=multiply(x,w)
    for(i=0;i<2*m;i++) p=p+'0'
    q=multiply(y,z)
    s1=sum(x,y)
    s2=sum(w,z)
    r=multiply(s1,s2)
    s=sum(r,-sum(p,q))
    for(i=0;i<m;i++) s=s+'0'
    return sum(sum(p,s),q)
```



```

multiply(array u,v) →  $T(n)$ 
    if(u=='0' or v=='0') return '0' →  $O(1)$ 
    n1=u.size()
    n2=v.size() →  $O(1)$ 
    n=max(n1,n2)
    if(n<threshold) return str(int(u)*int(v)) →  $T(1) = 1$ 
    m=n/2
    x,y=split(u)
    w,z=split(v) →  $O(1)$ 
    p=multiply(x,w) →  $T(\frac{n}{2})$ 
    for(i=0;i<2*m;i++) p=p+'0' →  $O(n)$ 
    q=multiply(y,z) →  $T(\frac{n}{2})$ 
    s1=sum(x,y) →  $O(n)$ 
    s2=sum(w,z)
    r=multiply(s1,s2) →  $T(\frac{n}{2})$ 
    s=sum(r,-sum(p,q)) →  $O(n)$ 
    for(i=0;i<m;i++) s=s+'0' →  $O(n)$ 
    return sum(sum(p,s),q) →  $O(n)$ 

```

$$T(n) = O(n) + 3T\left(\frac{n}{2}\right)$$

TIME COMPLEXITY-MASTER THEOREM

$$T(n) = O(n) + 3T\left(\frac{n}{2}\right)$$

$$a = 3, b = 2, k = 1, p = 0$$

$$T(n) = O(n^{\log_2 3}) = O(n^{1.58})$$

$$T(n) = a T\left(\frac{n}{b}\right) + \theta(n^k \log^p n)$$

$$\text{if } a > b^k, \text{ then } T(n) = \theta(n^{\log_b a})$$

ANY QUESTION??

HAVE A NICE DAY!